Huawei Access Node HSR-ACC-64 Optical Transport Failure SOP (ALARM_4078)

Node: HSR-ACC-64 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_4078

Description

The Huawei Access Node HSR-ACC-64, located at HSR Layout Access, is susceptible to Optical Transport Failures due to the inherent nature of SFP/XFP degradation, fiber cuts, and Rx/Tx power loss. This failure domain is exacerbated by the role of the Access Node in establishing and maintaining optical connections with remote nodes, thereby introducing a single point of failure. The Huawei Access Node, being a key component in the optical transport hierarchy, relies heavily on the SFP/XFP modules to transmit and receive optical signals. When these modules degrade or fail, it can lead to a cascade of failures downstream, ultimately resulting in the ALARM_4078 fault. Furthermore, the Access Node's reliance on fiber connections, which are prone to physical damage or cuts, adds to the likelihood of this failure domain. Therefore, it is essential to promptly identify and rectify the root cause of the ALARM_4078 fault to prevent further network disruptions.

Pre-requisite Checks

- Physical site visit to HSR Layout Access to assess the situation
- Verification of redundant link status to ensure failover capabilities are functioning correctly
- Confirmation of power supply to the Access Node

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low
Software or firmware issues	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical interface, Expected Output: Optical interface information, including SFP/XFP status and Rx/Tx power levels, Description: This command will provide insight into the current state of the optical interface, allowing for the identification of potential issues related to SFP/XFP degradation or fiber cuts.

Command: display optical fiber, Expected Output: Optical fiber information, including link status and physical connectivity, Description: This command will provide information about the physical state of the optical fiber, allowing for the identification of potential issues related to fiber cuts or physical damage.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP module, Command: replace optical interface , Description: This step involves replacing the SFP/XFP module on the Access Node to resolve any issues related to degradation or failure.

Step 2: Step: Repair or replace fiber cable, Command: repair optical fiber , Description: This step involves repairing or replacing the fiber cable to resolve any issues related to cuts or physical damage.

Step 3: Step: Update software or firmware, Command: upgrade system software, Description: This step involves updating the software or firmware on the Access Node to resolve any issues related to software or firmware related problems.

Post-Check Verification

```
Command: display optical interface, Expected Output: Optical interface information, including SFP/XFP status and Rx/Tx power levels, Description: This command will verify that the SFP/XFP module replacement or fiber repair has resolved the issue.

Command: display optical fiber, Expected Output: Optical fiber information, including link status and physical connectivity, Description: This command will verify that the fiber cable repair or replacement has resolved the issue.
```

Roll-back Procedures

- Step: Revert SFP/XFP module replacement, Command: undo replace optical interface , Description: This step involves reverting the SFP/XFP module replacement to its original state in case the new module does not resolve the issue.

- Step: Revert fiber cable repair or replacement, Command: undo repair optical fiber , Description: This step involves reverting the fiber cable repair or replacement to its original state in case the repair or replacement does not resolve the issue.